INCH-POUND

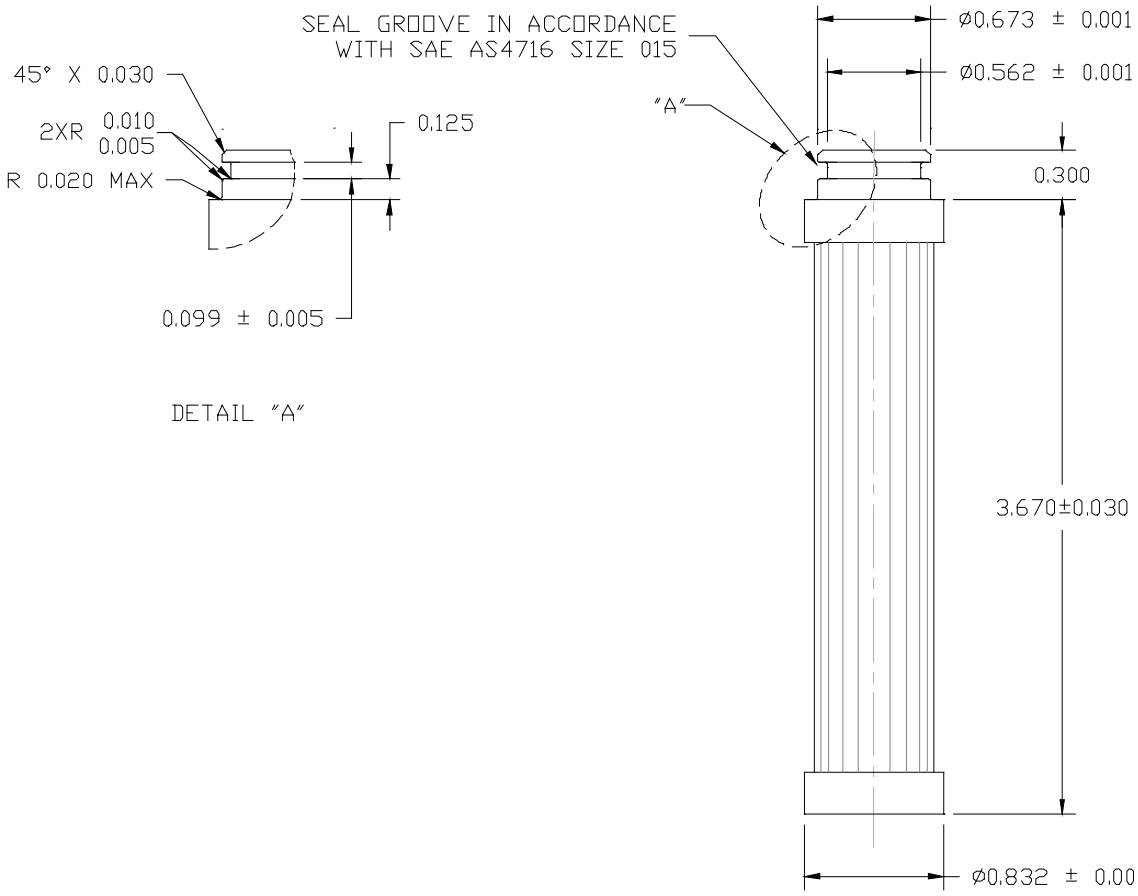
MIL-DTL-8815/31(AV)
16 October 2009

DETAIL SPECIFICATION SHEET

FILTER ELEMENTS, NON-CLEANABLE, FLUID PRESSURE, HYDRAULIC LINE, 3,000 PSI, 5 MICRON ABSOLUTE, - 65°F TO 275°F

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the filter element described herein shall consist of this document and the latest issue in effect of Specification MIL-F-8815.



Comments, suggestions, or questions on this document should be addressed to the Army Research Development and Engineering Command, Aviation and Missile Research and Engineering Center, Attn: AMSRD-AMR-SE-TD-ST, 5400 Fowler Road, Redstone Arsenal, AL 35895-5000, or email to streview@amrdec.army.mil.

AMSC N/A FSC 1650

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DRAWING NOTES:

1. Dimensions in inches unless otherwise specified.
2. Tolerances, decimals +/- .010, angles +/- 5°.
3. Finish all over 100% except as noted. Surface texture in accordance with ASA B46.1 – 1962 (ANSI) only to machined surfaces.
4. Material shall be in accordance with the procurement specification.
5. Drawing symbols per ASME Y14.5M – 1994.
6. Weight of filter element is 0.13 lbs dry max.
7. Rated flow is 6.0 gpm.

1. SCOPE.

1.1 Scope. This specification covers hydraulic filter elements for the AH-64 series helicopter. This specification describes:

- a multipass filtration performance test under cyclic flow and vibration conditions with continuous contaminant injection for hydraulic fluid power filter elements;
- a procedure for determining the contaminant capacity, particulate removal and differential pressure characteristics; and
- specific changes, revisions, and modifications to paragraphs of the latest issue in effect of MIL-F-8815

2. APPLICABLE DOCUMENTS.

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, Standards, and Handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-5606	Hydraulic Fluid, Fire Resistant, Petroleum Base, Aircraft, Missile and Ordnance
MIL-PRF-83282	Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Aircraft
MIL-PRF-87257	Hydraulic Fluid, Fire Resistant, Low Temperature, Synthetic Hydrocarbon Base, Aircraft and Missile

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(Copies of these documents are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094, or online at http://assist.daps.dla.mil/quicksearch/.)

2.3 Non-Government Publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

INTERNATIONAL ORGANIZATION FOR STANDARDS (ISO)

ISO 2941	Hydraulic fluid power – Filter elements – Verification of collapse/burst resistance
ISO 3968	Hydraulic fluid power – Filters - Evaluation of differential pressure versus flow characteristics
ISO 4406	Hydraulic Fluid Power – Fluids – Method For Coding the Level of Contaminants by Solid Particles
ISO 11171	Hydraulic fluid power – Calibration of automatic particle counters for liquids
ISO 11943	Hydraulic fluid power – On-line automatic particle counting systems for liquids – Method of calibration and validation
ISO 12103	Test Dust For Filter Evaluation
ISO 16889	Hydraulic fluid power – Filters – Multi-pass method for evaluating filtration performance of a filter element

(Copies of these documents are available from the American National Standards Institute, 1819 L Street, NW, Washington, DC 20036, or online at http://www.ansi.org).

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE) STANDARDS

SAE AS478	Identification Marking Methods
SAE ARP901	Bubble Point Test Method
SAE AS4059	Fluids, Hydraulic, Aerospace – Cleanliness Classification For

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SAE ARP4205	Aerospace Fluid Power-Hydraulic Filter Elements – Method for Evaluating Dynamic Efficiency With Cyclic Flow
SAE AS4716	Gland Design, O-Ring and Other Elastomeric Seals

(Copies of this document are available from Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096-0001, or online at http://www.sae.org.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-F-8815 and as modified herein.

3.2 Qualification. Items furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity (The US Army Aviation and Missile Command (AMCOM) Aviation Engineering Directorate (AED) Propulsion Engineering Division for listing in the Qualified Product Database (QPD) before contract award (see 4 and 6).

3.3 Part Number. The Part Number shall consist of the basic number of this specification sheet.

Example: M8815/31(AV)

4. VERIFICATION

4.1 General. The quality assurance requirements and revisions for the filter elements defined by this specification sheet shall be in accordance with MIL-F-8815 and as modified herein.

4.1.1 Particle Counter Validation Test Format. The validation test results shall be in accordance with (IAW) FIGURES B-1 through B-4. The acceptable sigma in FIGURES B-2 and B-3 can be obtained in accordance with paragraph 9.9 of ISO 11943.

4.1.2 Particle Counter Validation Test Format/Actual Test Results. The actual test results shall be IAW FIGURES B-5 through B-15.

4.2 Qualification – Re-Qualification**.** All Army AH-64 Apache Helicopter Hydraulic filter elements will require full qualification testing to this military specification sheet. To remain qualified, the manufacturer shall re-qualify every 3 years (even if the contract is awarded for a longer duration). Contract awards will be made only for such products as have been tested prior to the time set for opening of bids, been tested by an Army approved Laboratory and approved

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by AMCOM AED Propulsion Engineering Division for inclusion in the Army Aviation applicable Qualified Products Database (QPD). There will be no exceptions to this requirement. The attention of the suppliers is called to this requirement and manufacturers are urged to arrange to have the products that they propose to offer to the US Army tested for qualification, in order that they may be eligible to be awarded contracts or orders for the product covered by this specification. It is highly suggested that each filter manufacturer wishing to manufacture Army aircraft hydraulic filters under this specification provide the following information to the procuring activity (see 6.2.1).

4.3 ISO 16889 (see APPENDIX D for exceptions) Test Conditions. The filter elements furnished under this military specification sheet shall be subjected to the following test conditions:

Fluid Temperature:	160°F ± 15°F
Test Fluid:	MIL-PRF-83282 only
AH-64 Flow Range:	1.5 to 6.0 gallons per minute or 5.7 to 22.7 liters per minute
Flow Change Rate:	150 – 250 msec
Terminal Pressure:	300 psid
Test Dust:	ISO Medium Test Dust
Dilution:	None
Upstream Gravimetric:	3 mg/l

Particle Counting: 4, 5, 6, 7, 14 and 20 microns. Particle counting shall be synchronized with flow changes as depicted in FIGURE A-3. At least one of the data points for upstream and downstream counts shall be taken after 7 seconds of changes in flow every flow cycle. Examples: Counting shall start after 7 seconds of flow change from low to high flow and one other after high to low flow. Do not take particle count readings between flow changes. Counting shall continue as depicted in FIGURE A-3 for numerous specific events, as applicable and recorded in FIGURES B8 and B9.

Cyclic Frequency ~0.004 Hz (see profile of particle count synchronization) shall be as shown in FIGURE A-3.

Mechanical Vibration: The mechanical vibration levels consist of a sinusoidal profile with frequency set at 38 Hz and peak acceleration set at 5g's. The Vibration shall be maintained during the Dynamic Efficiency Tests.

Marking: Each test article shall be laser marked or metal stamped with a minimum of 0.040 inch high characters or Vibro Etched for permanent unique identification IAW SAE AS478. Orientation is optional. Each test article shall be marked with the following information:

Part Number/CAGE
Vendor Part Number/Lot No

4.4 Responsibility for Qualification – Re-qualification Requirement**.** The activity responsible for this specification and qualification and re-qualification requirement is US Army Aviation and Missile Command, Aviation Engineering Directorate, Propulsion Division, RDMR-AEP, Building 4488, Redstone Arsenal, Alabama 35898-5000. Based on the suggested requirement of

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the contractor providing recommended manufacturing planning and implementing a Change Control Board (CCB) for any and all changes, the requirement of re-qualification such as a lot change due to filter element material change, contract completion and break in delivery or the 3-year re-qualification requirement may be renegotiated. If manufacturing planning is not provided, the contractor will be removed from the qualified vendors list and will have to re-qualify per paragraph 4.2 herein.

4.4.1 Quality Conformance. The quality conformance testing consists of the following tests: The Quality Conformance Testing outlined in MIL-F-8815 Table II will be followed except for the Degree of Filtration paragraph 4.7.2.4. Validation Test (if older than 6 months, the Validation Test shall be repeated), and three Dynamic Efficiency Tests on pre-conditioned filter elements shall be part of this requirement. Data shall be reported in Appendix B for this Validation and Conformance Test.

4.4.2 Quality Conformance Testing. The following changes constitute the quality conformance testing: Media, New Materials, Manufacturing Processes, Re-opening of the contract, and Company Buyout.

4.5 Filter Housing Approval**.** The filter housing, which shall house the filter element, shall be approved by the Government prior to qualification of the filter element undergoing development and tests.

4.6 Filter Elements**.** Qualification test samples of filter elements (non-cleanable) shall consist of nine elements upon which qualification is desired. Manufacturer's qualification tests shall be performed on these test samples in accordance with this military specification sheet as defined herein.

4.7 Retention**.** The retention of qualification shall consist of periodic verification to determine compliance of the qualified filter assembly and filter element with the requirements of this specification and applicable specification sheets. Verification shall be conducted at intervals not exceeding 3 years.

4.8 Individual Inspection/Examination**.** The individual inspection, specified in this specification shall be conducted on each filter element as shown in the FIGURE A-1. Each filter shall be carefully examined to determine conformance to the requirements of this specification and MIL-F-8815 for design, weight, workmanship, marking, conformance to applicable standards, government and manufacturer's drawings, specifications, and standards for visible defects. Any filter element containing a defect shall be rejected.

4.9 Test Fluid.

4.9.1 Hydraulic Test Fluid**.** Unless otherwise specified, the hydraulic fluid used for all tests shall conform to MIL-PRF-5606 (with no free water), or MIL-PRF-83282 (with no free water), or MIL-PRF-87257 (with no free water). These test fluids may be used for quality conformance tests as specified in this military specification sheet.

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4.9.2 Test Fluid Cleanliness**.** For quality conformance tests and critical qualification tests, the test fluid shall be pre-cleaned to a cleanliness level equal to ISO 13/10/7 or SAE AS4059 (NAS) Class 1 or less.

4.10 Immersion/Thermal Cycling**.** All filter elements shall be immersed in hydraulic fluid for a total period of 192 hours. Filter elements shall be immersed in MIL-PRF-83282 or MIL-PRF-87257 at a fluid temperature of 275°F for 24 continuous hours followed immediately (within 10 minutes) by soaking at -65°F using MIL-PRF-5606, MIL-PRF-87257, or MIL-PRF-83282. The filter element shall immediately (within 10 minutes) be placed back into the hot fluid bath of 275°F for 24 continuous hours followed again by the cold -65°F using MIL-PRF-5606, MIL-PRF-87257, or MIL-PRF-83282. This test cycle (hot/cold) shall be repeated four times, for a total immersion test time of 96 hours of the hot fluid bath and a total of 96 hours of the cold immersion bath on the test filter. Filters may remain immersed in test fluid at room temperature until the filters are ready for rinsing and visual inspection. This test will require two (2) environmental chambers to perform.

4.10.1 Post Immersion Visual Examination. Visually examine test article for evidence of deterioration after the Immersion Test above. Document the condition of the test article after this testing. Take photographs, including close-ups of filter media after testing.

4.10.2 Post Immersion Storage. After the Immersion Test and Post Immersion Visual Examination is completed, the filter shall remain in one of the three hydraulic fluids identified in 4.9.1 at room temperature until further tests are completed. If testing is suspended for any reason, this paragraph applies to keep the filter immersed in hydraulic fluid for further tests.

4.11 Test Dust, Source of Supply**.** All ISO Medium Test Dust or ISO 12103-1A3 Test Dust shall be used for tests and shall be obtained from Power Technology Inc., 14331 Ewing Ave., South Burnsville, MN 55306 or other sources approved per ISO 12103.

4.12 Bubble Point Qualification**.** Prior to performing qualification tests, all filter elements shall be Bubble Point tested per SAE ARP901 to determine the minimum allowable initial Bubble Point in accordance with paragraph 4.12.2 below. After physical examination, clean the filter element, ensuring no fluid exists. Then the filter element shall be tested in accordance with SAE ARP901. This test shall again be accomplished within a period of 15 - 30 minutes after the immersion test. Elements removed from storage after immersion test may be gently rinsed with isopropyl alcohol and/or test fluid prior to initiation of bubble test and visual inspection.

4.12.1 Bubble Point After Performance Test**.** Where specified in the Qualification Test Procedure of this specification, a Bubble Point test shall be conducted per 4.13 below. This test shall be performed after every Immersion Test and Cold Start Test. The Bubble Point value shall not be less than the minimum allowable initial Bubble Point (see 4.12.2 below). Filter elements shall be rotated 360° about its longitudinal axis at each increment of air pressure so that the entire area is observed for appearance of the first bubble stream.

4.12.2 Minimum Allowable Initial Bubble Point**.** The Dynamic Efficiency Test of 4.15 shall be performed on three filter elements with the lowest Bubble Point. The Collapse Test of 4.16 shall be performed on two filter elements with the highest Bubble Points. If the elements pass the minimum Dynamic Efficiency Test of 4.15, then the lowest Bubble Point element tested after

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Cold Start Test shall be considered the minimum allowable bubble point. The minimum allowable initial Bubble Point shall be documented. Bubble points measured after the Cold Start Testing on clean dried elements forms the basis for ranking of the bubble point.

4.13 Quality Conformance Bubble Point**.** The cleaned (rinsed) filter element shall be installed in a setup as specified in SAE ARP901, paragraph 5. Air pressure shall be preset to 0.1 inch of water less than the minimum initial bubble point determined in 4.12. Since the minimum bubble point value will differ for various filtration media, the bubble point value shall be previously approved and obtained in writing with a qualification approval letter from the US Army, AED Propulsion Division for the particular filter element under test.

4.14 Cold Start**.** The filter elements shall be installed in the housing filled with MIL-PRF-5606, MIL-PRF-87257, or MIL-PRF-83282, and subjected to a temperature of -65 +0 /-5°F for 4 hours in a test setup similar to FIGURE C-2 or a pre-approved equivalent test facility. The AMCOM AED Propulsion Engineering Division shall be notified at least 30 days in advance of the test facility recommended for this test. Only AMCOM AED Propulsion Engineering Division approved test facilities will perform this test. The filter element shall then be subjected to 10 flow cycles at a differential pressure of 3,000 psid across the Army approved assembly without restriction in the housing with fluid held at -65 +0 /-5°F. Each flow cycle shall be 15 +0 / -1 second in duration where the outset of the cycle shall increase rapidly to a differential pressure of 3,000 psi and be maintained for duration of 7.5 +1.0 / -0.0 seconds and then be reduced rapidly to 0 – 100 psi for the remainder of the cycle duration, see FIGURE A-2. No bypass of fluid shall be allowed during this test. There shall be no evidence of filter media or structural damage of any kind for each element tested. The same filter elements shall then be subjected to satisfactory completion of subsequent tests listed in the attached Qualification Test Procedure, see FIGURE A-1.

4.15 Dynamic Efficiency**.** Three filter elements shall undergo the Dynamic Efficiency Test as specified in ISO 16889 (paragraph 11 Filter Performance Test) (see APPENDIX D for exceptions) in addition to 4.1.1, 4.1.2, and 4.3 listed herein. Figure B.1 of the ISO 16889 shall be replaced with FIGURE C-1 herein for use with this military specification sheet (see Appendix D for additional exceptions to ISO 16889). Prior to performing this test, the filter company wishing to qualify under this requirement, must notify and receive approval in writing, from the US Army Aviation and Missile Command, Aviation Engineering Directorate, Propulsion Division RDMR-AEP, Building 4488, Redstone Arsenal, Alabama 35898-5000. This requirement covers the validation of the contaminant injection system; to 3 mg/l as specified in the ISO 16889 (see APPENDIX D for exceptions) paragraphs 8.2 and Table-4 of this specification. Calibration records and validation test results shall be IAW ISO 16889 (see Appendix D for exceptions) paragraph 8.1, modified to perform validation of the filter test system following the modified test parameters of 4.3. Filter elements provided under this military specification sheet shall meet the following minimum performance requirements at 6 microns (c) at 300 psid terminal pressure across the test element:

Grand Average ($\alpha_{6(c)}$):	80
AFR (High to $Low_{6(c)}$):	35
AFR (Low to $High_{6(c)}$):	35
Retained Capacity:	0.5 grams

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The contractor shall provide the Government not less than 30 days advance notice of this test to allow the Government to witness both test validation of the test system and validation of the contaminant injection system. Documentation shall be IAW ISO 11171 and ISO 11943 Table 3. At cyclic conditions during this test, documentation shall cover the Flow Profile – Particle Synchronization as depicted in the FIGURE A-3 of this military specification sheet. This documentation shall cover both high to low and then low to high particle count cyclic flow readings from the 1.5 to 6.0 gallons per minute rated flow.

4.16 Collapse/Burst Resistance**.** The filter element shall withstand a differential pressure of 4,500 psid in accordance with ISO 2941, except that paragraph 6.7 of ISO 2941 shall be as follows: Inject into the system 0.1 gram of ISO Medium Test Dust or ISO Fine Test Dust in a continuous fashion every 4 minutes while maintaining the rated nominal flow and test temperatures.

4.17 Pressure Drop**.** The filter elements shall undergo a differential pressure test measuring the differential pressure upstream and downstream of the filter under test using a differential pressure transducer or two gauges pressure transducers as specified in ISO 3968, at 100°F ± 5°F using MIL-PRF-83282 only. The pressure differential between the Pre-Cold Start Test and the Post-Cold Start Test shall not exceed 25%. The pre-cold start pressure drop across the element at 100°F, 6 gpm flow rate shall not be greater than 23 psid.

4.18 Flow Fatigue**.** The filter element shall be installed in a Government-approved, non-bypass type filter housing (see 4.5). The filter shall be subjected to pressure-flow cycles with hydraulic fluid at a minimum of 30% of rated flow and at 275°F ± 15°F. A cycle shall consist of increasing the differential pressure across the filter element from zero to the maximum specified and back to zero by first increasing, then decreasing, the flow through the test filter element, which has been loaded with ISO Medium Test Dust or ISO Fine Test Dust. Peak pressure shall be reached within 25% of the cycle and held for at least 50% of each cycle, see FIGURE A-4. The bubble point test following the flow fatigue shall not be less than the minimum bubble point on the elements that passed Dynamic Efficiency Test specifications per 4.12.2. It is highly encouraged to take down stream fluid samples during the last 100 cycles and report the data in terms of particle counts ISO 4406 or SAE AS4059. The report will not be used as a failure criterion but for information only. The filter element shall withstand the following flow fatigue pressure cycles at a cycling rate of 100 cycles per minute maximum**:**

90,000 cycles at 20 psid
8,000 cycles at 50 psid
2,000 cycles at 100 psid
500 cycles at 3,000 psid

4.19 Media Migration**.** The filter elements shall undergo the Media Migration Test as specified in MIL-F-8815 except as reported in Table I below.

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TABLE I. Media Migration Contaminants Analysis.

ACFT	Flow Rate	Migration Analysis Value Maximum Allowable	
	(GPM)	(Grams)	
		After Cold Start	After Vibration
AH-64	6.0	0.0035	0.0016

5. PACKAGING.

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (sees MIL-F-8815, paragraph 4.8). When packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging authority.

6. NOTES.

6.1 Intended use. This slash sheet is to be used in conjunction with MIL-F-8815 to procure non-cleanable, fluid pressure, hydraulic filter elements for AH-64 aircraft.

6.2.1 Manufacturing Planning. All manufacturing, assembly, and inspection points shall be controlled by a detailed written procedure outlining each step or parameter of the filter manufacturing process along with any materials, tooling equipment, environmental control, and operator certification required that leads to the specific filter production. Plans shall clearly identify all critical materials, processes and procedures and will include identification in accordance with contractor procedures, as to its particular revision. All process plans shall clearly define sequences of operation, machine type, and accept/reject limits for the specific process or operation. Any critical processes not easily verified by subsequent inspection shall be clearly defined in the process operation with tolerances.

6.2.2 Frozen Planning Requirements. The contractor is responsible for developing the manufacturing planning document described above. Review and control of these plans will be the responsibility of the Contractor's Control Board (CCB) consisting of qualified personnel equipped with adequate resources to assure development of complete, reliable, and traceable documentation. Aircraft filters manufactured utilizing these plans shall meet all contractual requirements of this Military Specification Sheet. Once frozen, plans shall remain frozen throughout the existing contract and all subsequent contracts for the filter unless changes to the planning are made in accordance with this Military Specification Sheet.

6.2.3 Changes to Frozen Planning. The portion of the frozen manufacturing plan pertaining to a critical process or material during the construction of the filter element shall not be changed

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without prior CCB recommendation and justification to AMCOM, AED Engineering, and receipt of approval by the Procurement Contracting Officer (PCO). Changes not affecting critical processes or material or occurring as the result of an AMCOM approved Engineering Change Proposal require CCB approval only. All changes to frozen planning affecting critical process or material will be submitted to AMCOM AED Engineering. When the filter element critical process or material is produced by a subcontractor, the planning shall be reviewed and approved by the contractor, and be subjected to the same restriction as above.

6.2.4. Qualification Test Report**.** The qualification test report should include the following:

a. The validation test results as described in 4.1.1.

b. The actual test results as described in 4.1.2.

c. The test reports, submitted as specified and in full knowledge of the US Army, as defined in 4.4.

d. The minimum allowable initial Bubble Point defined in 4.12, which should also be included in the filter element drawing or the applicable production Acceptance Test Procedure (ATP).

e. A graph showing the flow cycle-pressure time and oil temperature as defined in 4.14.

f. A graph showing the test dust added versus pressure differential as defined in 4.16.

g. Data to substantiate the flow fatigue test operating conditions of 4.18.

6.3 Subject term (key word) listing.

Capacity, contaminant
Helicopter, AH-64
Performance test, multi-pass filtration
Removal, particulate

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APPENDIX A

MIL-DTL-8815/31(AV)
US ARMY AVIATION
QUALIFICATION TEST PROCEDURE

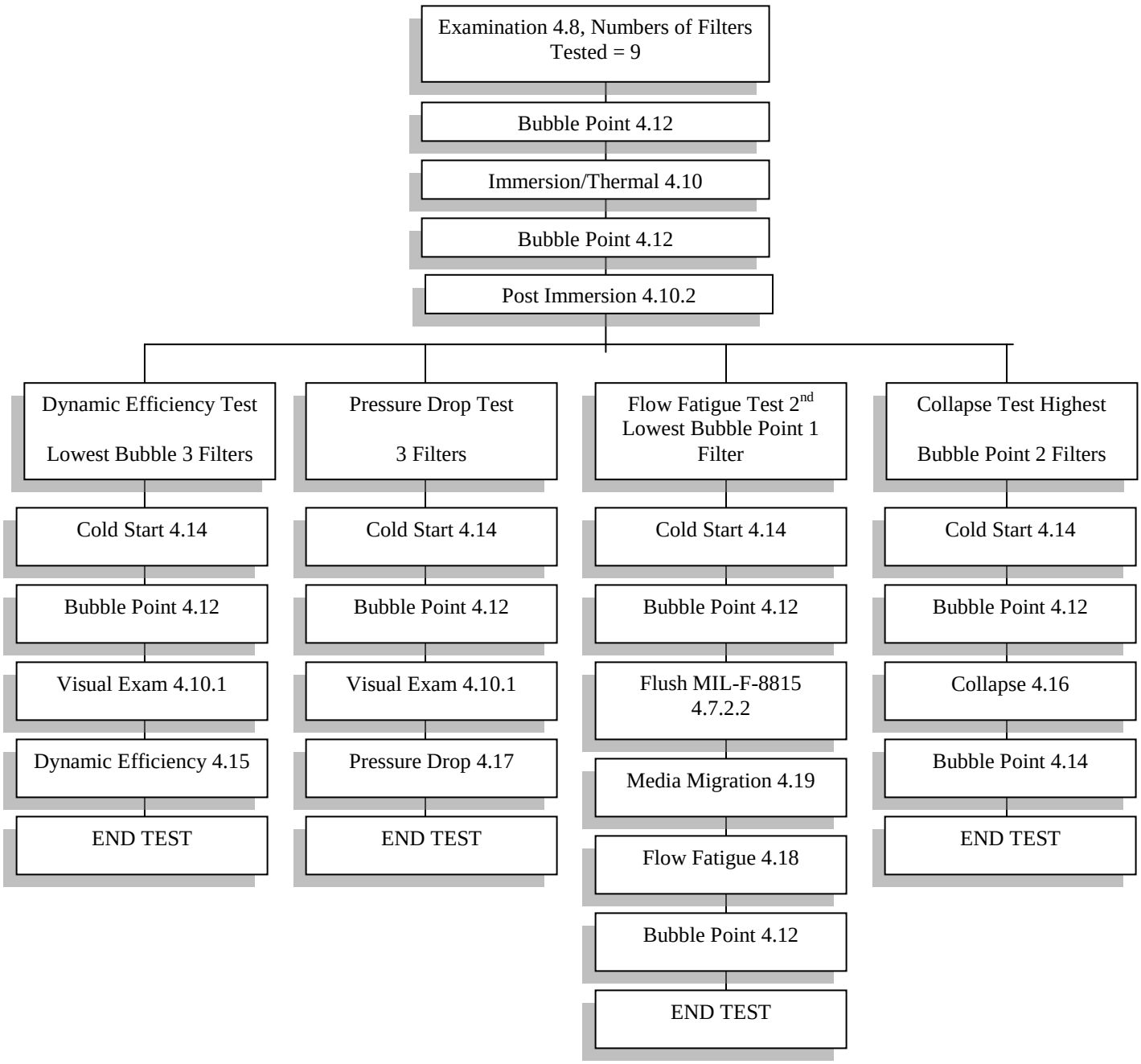
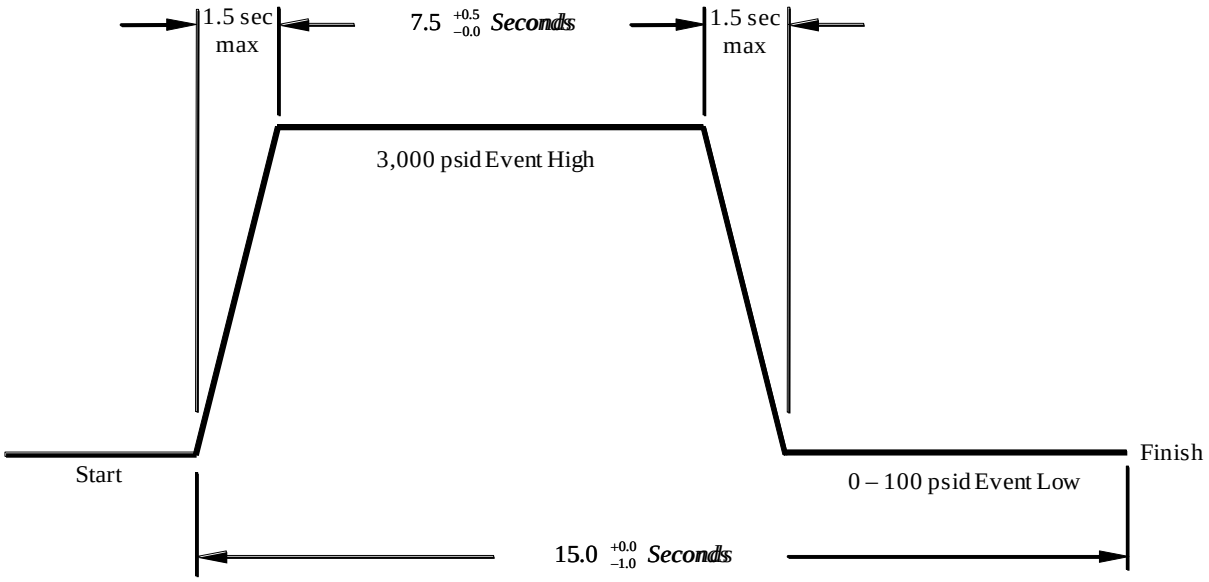


FIGURE A-1. MIL-DTL-8815/31(AV) US Army Aviation Qualification Test Procedure.

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APPENDIX A



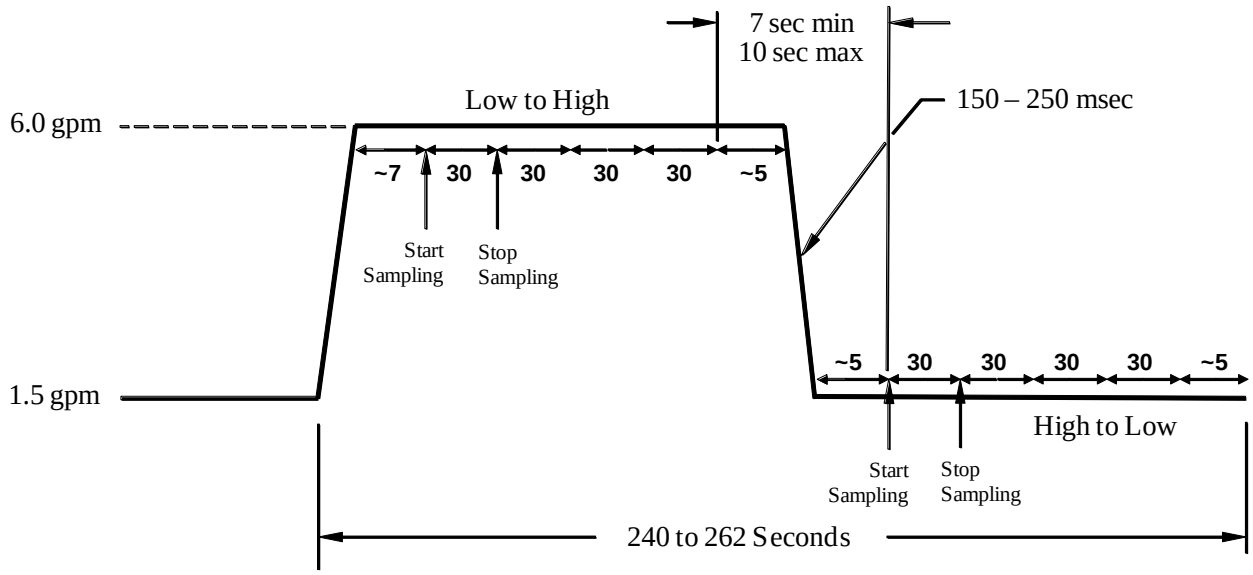
Cold Start Test Cycle To Be Repeated 10 Times

FIGURE A-2. Cold Start Test Cycle.

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APPENDIX A

Algorithms used for calculation of Filtration Ratios using cyclic multipass test with vibration using synchronized particle counting. Counting shall start within 7 seconds after change of flow from low to high flow and one other after high to low flow. Do not take particle count readings between flow changes.



Note: Delay time between count periods is a function of particle counter design – can be 0 – 5 seconds.

Average Filtration Ratio (FR) for High to Low (H/L), Low to High (L/H) Events, and Grand Average:

$$\text{AFR (High to Low}_{(c)}) = \frac{\sum \text{(Upstream Counts) High to Low Event}}{\sum \text{(Downstream Counts) High to Low Event}}$$

$$\text{AFR (Low to High}_{(c)}) = \frac{\sum \text{(Upstream Counts) Low to High Event}}{\sum \text{(Downstream Counts) Low to High Event}}$$

$$\text{Grand Average Filtration Ratio, } \alpha_{x(c)} = \frac{\sum_{k=1}^{10} \bar{N}_{u,x,t}}{\sum_{k=1}^{10} \bar{N}_{d,x,t}}\text{, according to the ISO 16889}$$

(see APPENDIX D for exceptions)

FIGURE A-3. Dynamic Efficiency Test Profile – Particle Synchronization.

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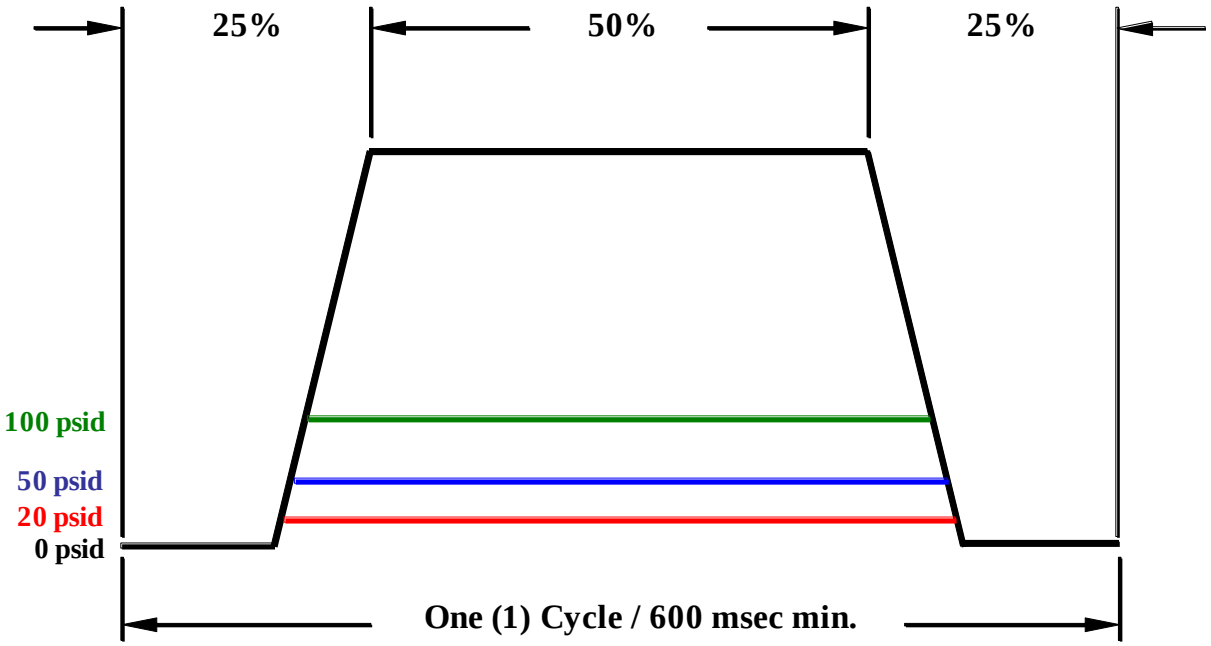


FIGURE A-4. Flow Fatigue Test Profile.

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APPENDIX B

Validation Report	
Test Procedure Name	Aircraft
Test Number	Housing ID
Test Date	NSN No.
Start Test Time	Project Name
Operator	Test Type
Terminal Pressure	Report Date
Test Temperature	Test Volume
High or Main Flow	Dust Lot Number
Low Flow	Inj Sys Vol
Cyclic High Flow Interval	Inj Sys Mass
Cyclic Flow Flow Interval	Particle Count
	Sample Time
Remarks	

FIGURE B-1. Validation Report.

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APPENDIX B

Test Report - Calibration Sheet

Table 1 - Secondary calibration dust data sheet (Upstream)

Test Number:	Test Date:	Operator:
ISO MTD lot no:	Concentration:	Sensor Model:
Particle counter no:	Particle count volume:	ISO 11171 primary cal. date:
Sensor SN:	Particle counter Model:	Test Temperature:

Count Number	Time Seconds	d>4 µm(c)	d>5 µm(c)	d>6 µm(c)	d>7 µm(c)	d>10 µm(c)	d>12 µm(c)	d>14 µm(c)	d>15 µm(c)	d>20 µm(c)	d>21 µm(c)	d>25 µm(c)	d>30 µm(c)
Count 1	60												
Count 2	180												
Count 3	300												
Count 4	420												
Count 5	540												
Count 6	660												
Count 7	780												
Count 8	900												
Count 9	1020												
Count 10	1140												
Count 11	1260												
Count 12	1380												
Count 13	1500												
Count 14	1621												
Count 15	1741												
Count 16	1861												
Count 17	1981												
Count 18	2101												
Count 19	2221												
Count 20	2341												
Count 21	2461												
Count 22	2581												
Count 23	2701												
Count 24	2821												
Count 25	2941												
Count 26	3061												
Count 27	3181												
Count 28	3301												
Count 29	3422												
Count 30	3542												
Mean													
Sigma													
Accep. Sigma													
Pass/Fail													

FIGURE B-2. Test report – calibration sheet.

Note: Refer to ISO 11943 for computing sigma and acceptable sigma

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APPENDIX B

Test Report - Calibration Sheet

Table 2 - Secondary calibration dust data sheet (Downstream)

Test Number:	Test Date:	Operator:
ISO MTD lot no:	Concentration:	Sensor Model:
Particle counter no:	Particle count volume:	ISO 11171 primary cal. date:
Sensor SN:	Particle counter Model:	Test Temperature:

Count Number	Time Seconds	d>4 µm(c)	d>5 µm(c)	d>6 µm(c)	d>7 µm(c)	d>10 µm(c)	d>12 µm(c)	d>14 µm(c)	d>15 µm(c)	d>20 µm(c)	d>21 µm(c)	d>25 µm(c)	d>30 µm(c)
Count 1	60												
Count 2	180												
Count 3	300												
Count 4	420												
Count 5	540												
Count 6	660												
Count 7	780												
Count 8	900												
Count 9	1020												
Count 10	1140												
Count 11	1260												
Count 12	1380												
Count 13	1500												
Count 14	1621												
Count 15	1741												
Count 16	1861												
Count 17	1981												
Count 18	2101												
Count 19	2221												
Count 20	2341												
Count 21	2461												
Count 22	2581												
Count 23	2701												
Count 24	2821												
Count 25	2941												
Count 26	3061												
Count 27	3181												
Count 28	3301												
Count 29	3422												
Count 30	3542												
Mean													
Sigma													
Accep. Sigma													
Pass/Fail													

FIGURE B-3. Test report – calibration sheet.

Note: Refer to ISO 11943 for computing sigma and acceptable sigma

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APPENDIX B

Table 3 - Particle counts per mL for ISO MTD (Upstream) at 1 mg/liter

Particle size µm(c)	Primary calibration Contaminant**	Avg of 30 * counts for ISO MTD lot(Table 1)	Calibration Limits	Difference Between Column 2 and Column 3	Acceptance Limits (1.3 x Column 4)	Pass or Fail
>4	2176.7		254.3		330.6	
>5	1330		167.3		217.5	
>6	855		114.9		149.4	
>7	561.7		80.4		104.5	
>10	183		31		40.3	
>12	100		18.5		24.1	
>14	60.8		12.1		15.8	
>15	51.5		10.5		13.7	
>20	18.3		4.4		5.7	
>30	3		0.9		1.2	

Table 4 - Particle counts per mL for ISO MTD (Downstream) at 1 mg/liter

Particle size µm(c)	Primary calibration Contaminant**	Avg .of 30 * counts for ISO MTD lot(Table 2)	Calibration Limits	Difference Between Column 2 and Column 3	Acceptance Limits 1.3 x Column 4)	Pass or Fail
>4	2176.7		254.3		330.6	
>5	1330		167.3		217.5	
>6	855		114.9		149.4	
>7	561.7		80.4		104.5	
>10	183		31		40.3	
>12	100		18.5		24.1	
>14	60.8		12.1		15.8	
>15	51.5		10.5		13.7	
>20	18.3		4.4		5.7	
>30	3		0.9		1.2	

Table 5 - Particle counts per mL for ISO MTD (Between Upstream and Downstream) at 1 mg/liter

Particle size µm(c)	Primary calibration Contaminant**	Avg. of 30* counts for ISO MTD lot Upstream (Table 1)	Avg. of 30* counts for ISO MTD lot Downstream (Table 2)	Difference Between Counters	Variation allowed between counters***	Pass or Fail
>4	2176.7				109.4	
>5	1330				67.1	
>6	855				43.4	
>7	561.7				28.7	
>10	183				9.8	
>12	100				5.6	
>14	60.8				3.6	
>15	51.5				3.2	
>20	18.3				1.5	
>30	3				0.8	

* average of minimum of 30 counts of actual lot of dust used when measured on -line on test stand
** average of primary ISO Med calibration contaminant per ISO 16889 table (1 mg/ml)
*** variation calculated from ISO 11943 para 9.17

FIGURE B-4. Tables used for particle counts per mL for ISO MTD.

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APPENDIX B

Test Results

Test Procedure Name	Aircraft
Test Number	Housing ID
Test Date	NSN No.
Start Test Time	Vendor Number
End Test Time	Project Name
Operator	Test Type
Terminal Pressure	Report Date
Test Temperature	Test Volume
High or Main Flow	Dust Lot Number
Low Flow	Inj Sys Vol
Cyclic High Flow Interval	Inj Sys Mass
Cyclic Low Flow Interval	Particle Count Sample Time
Remarks	

FIGURE B-5. Test Results.

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APPENDIX B

Test Report

Test Number	Test Date	Operator

FILTER AND ELEMENT IDENTIFICATION

Element ID:	Housing ID:
Spin on:	Minimum element bubble point:

OPERATION CONDITIONS

Test Fluid

Type: Ref: Batch No:

Viscosity at the test temperature (mm^2/s): Test temperature range (°C):

Antistatic: Type: Conductivity (pS/m):

Test Contaminant

Type: Batch No:

Test System

Flow rate q_{max} (L/min): Initial volume (L):

Base upstream concentration G_b (mg/L): Final volume (L):

Injection System

Injection parameter	*Initial*	*Final*	*Average injection parameters*	
System volume (L):			Injection flow q_i (L/min):	
Concentration (mg/L):			Concentration G_i (mg/L):	

Counting System	*Counter and sensor*	*Flow Rate (mL/min):*	*Dilution ratio*
Upstream			
Downstream			

Counter calibration Method: Date:

TEST RESULTS

Element Integrity

Bubble point to ISO 2942 Wetting fluid:

Differential Pressure

Initial ass'y, Δp (kPa):

Final element, Δp (kPa):

Retention Capacity

ISO MTD mass injected M_i (g): ISO MTD retained capacity C_R (g):

80% upstream concentration G_{80} (mg/L):

Filtration Ratio $\alpha_{x(c)}$

Average Filtration Ratio						
Particle size, μm(c)						

FIGURE B-6. Test Report

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APPENDIX B.

Differential Pressure Versus Containment Added

Time Interval	Test Time (min)	Element Δp (kPa)	Injected mass (g)	Time Interval	Test Time (min)	Element Δp (kPa)	Injected mass (g)
0%				60%			
10%				70%			
20%				80%			
30%				90%			
40%				100%			
50%							

Grand Average Filtration ($\alpha_{(c)}$) Ratio Report

Time Interval	d>4 µm(c)	ratio α	d>5 µm(c)	ratio α	d>6 µm(c)	ratio α	d>7 µm(c)	ratio α	d>10 µm(c)	ratio α
Initial up										
10% Up										
Down										
20% Up										
Down										
30% Up										
Down										
40% Up										
Down										
50% Up										
Down										
60% Up										
Down										
70% Up										
Down										
80% Up										
Down										
90% Up										
Down										
100% Up										
Down										
Avg Up										
Avg Down										

FIGURE B-7. Grand Average Filtration ($\alpha_{(c)}$) Report

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APPENDIX B

Dynamic Efficiency Test Report Average Filtration Ratio (AFR) low to high

Operating Conditions

Record Time to cleanup: to ISO code level on preconditioned element – ISO 13/10/7 or below:

Cycle High Flow Interval:

Cycle Low Flow Interval:

Flow Rate: Low

High

Final Δp element:

Operating Temperature:

Vibration:

Flow Rise rate:

Retention Capacity

ISO MTD mass injected M_i (g):

ISO MTD retained capacity C_R (g):

80% upstream concentration G_{80} (mg/L):

Filtration Ratio

Average filtration ratio						
Particle size, µm(c)						

Particle counts per mL and Filtration Ratio – Average Filtration Ratio (AFR) low to high

Time Interval	d>4 µm(c)	ratio α	d>5 µm(c)	ratio α	d>6 µm(c)	ratio α	d>7 µm(c)	ratio α	d>10 µm(c)	ratio α
Initial Up										
1 event Up										
Down										
2 event Up										
Down										
3 event Up										
Down										
4 event Up										
Down										
5 event Up										
Down										
6 event Up										
Down										
7 event Up										
Down										
Avg. Up										
Avg. Down										

FIGURE B-8. Dynamic Efficiency Test Report – Average Filtration Ratio (AFR) Low to High.

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APPENDIX B

Dynamic Efficiency Test Report Average Filtration Ratio (AFR) high to low

Operating Conditions
Record Time to cleanup: to ISO code level on preconditioned element – ISO 13/10/7 or below:
Cycle High Flow Interval:
Cycle Low Flow Interval:
Flow Rate: Low Final Δp element:
High
Operating Temperature:
Vibration:
Flow Rise rate:

Retention Capacity

ISO MTD mass injected M_i (g): ISO MTD retained capacity C_R (g):

80% upstream concentration G_{80} (mg/L):

Filtration Ratio

Average filtration ratio						
Particle size, µm(c)						

Particle counts per mL and Filtration Ratio – Average Filtration Ratio (AFR) high to low

Time Interval	d>4 µm(c)	ratio α	d>5 µm(c)	ratio α	d>6 µm(c)	ratio α	d>7 µm(c)	ratio α	d>10 µm(c)	ratio α
Initial Up										
1 event Up										
Down										
2 event Up										
Down										
3 event Up										
Down										
4 event Up										
Down										
5 event Up										
Down										
6 event Up										
Down										
Avg. Up										
Avg. Down										

FIGURE B-9. Dynamic Efficiency Test Report – Average Filtration Ratio (AFR) High to Low

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APPENDIX B

Cleanliness downstream levels of the filter prior to start of the test – particle counts per mL

Time Seconds	>4 μm(c)		>6 μm(c)		>14 μm(c)		>4 μm(c) Threshold level before starting the test	>6 μm(c) Threshold level before starting the test	>14 μm(c) Threshold level before starting the test
	Up Stream	Down Stream	Up Stream	Down Stream	Up Stream	Down Stream			

FIGURE B-10. Cleanliness Downstream Levels

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APPENDIX B

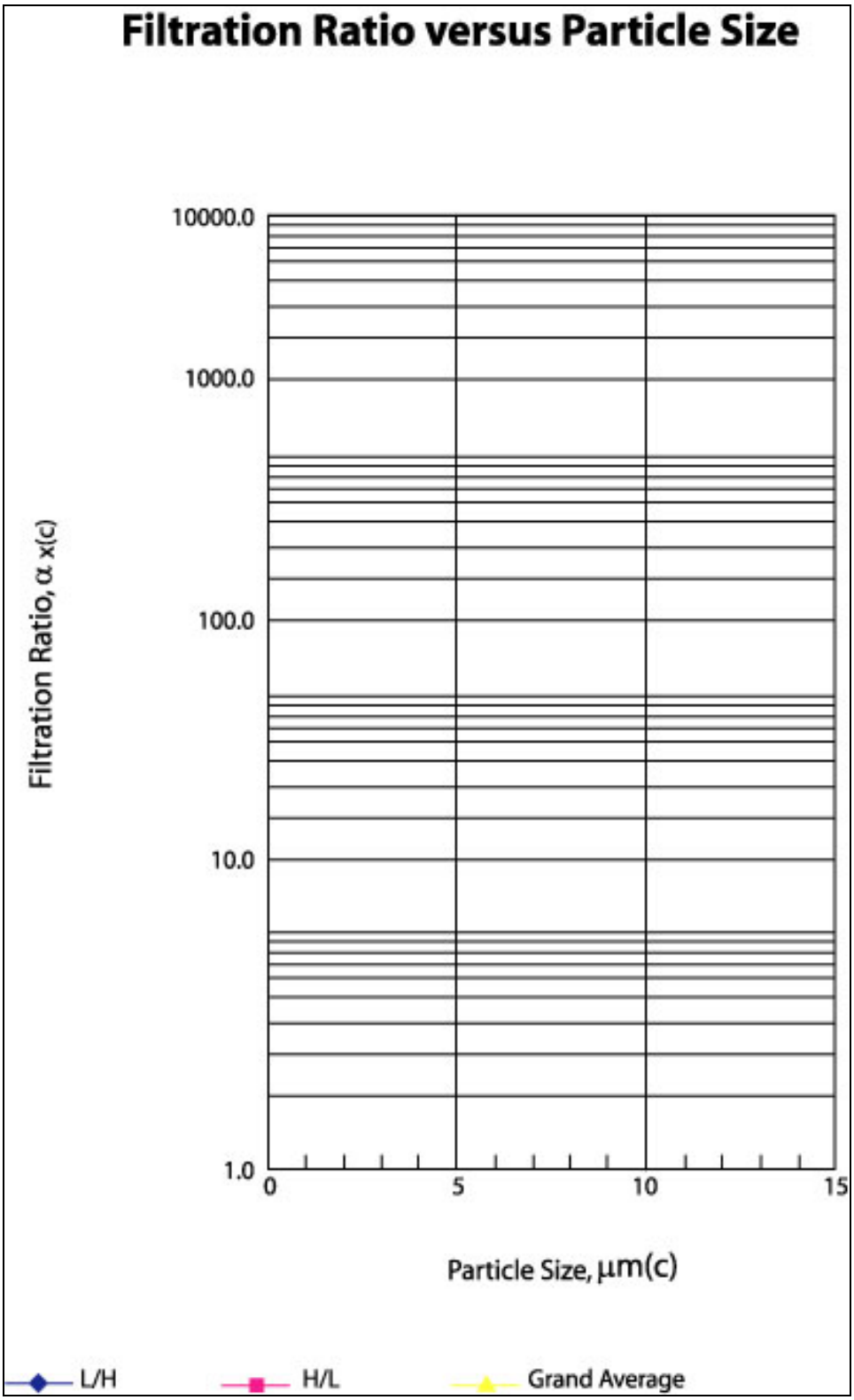


FIGURE B-11. Filtration Ration Versus Particle Size

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APPENDIX B

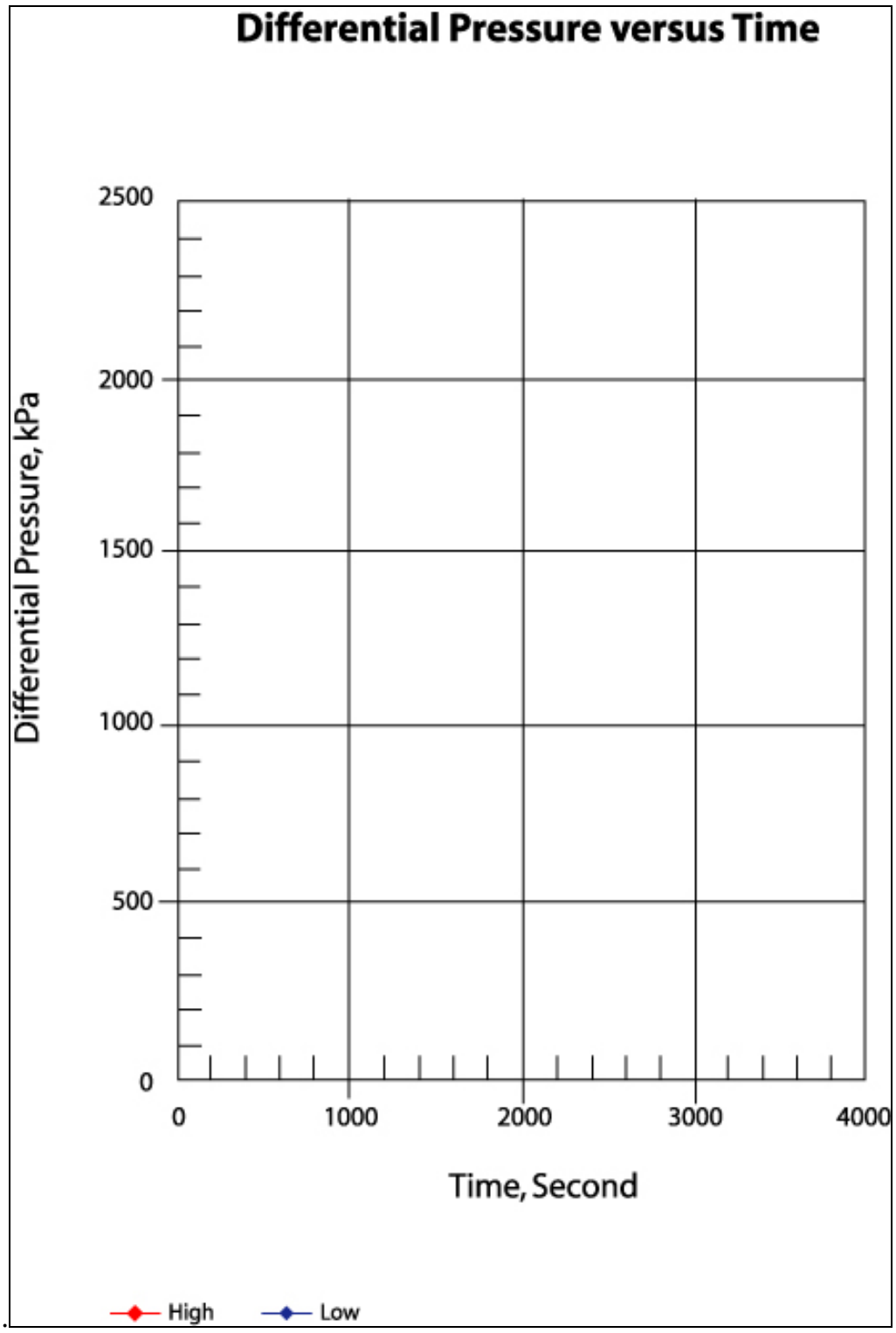


FIGURE B-12. Differential Pressure Versus Time

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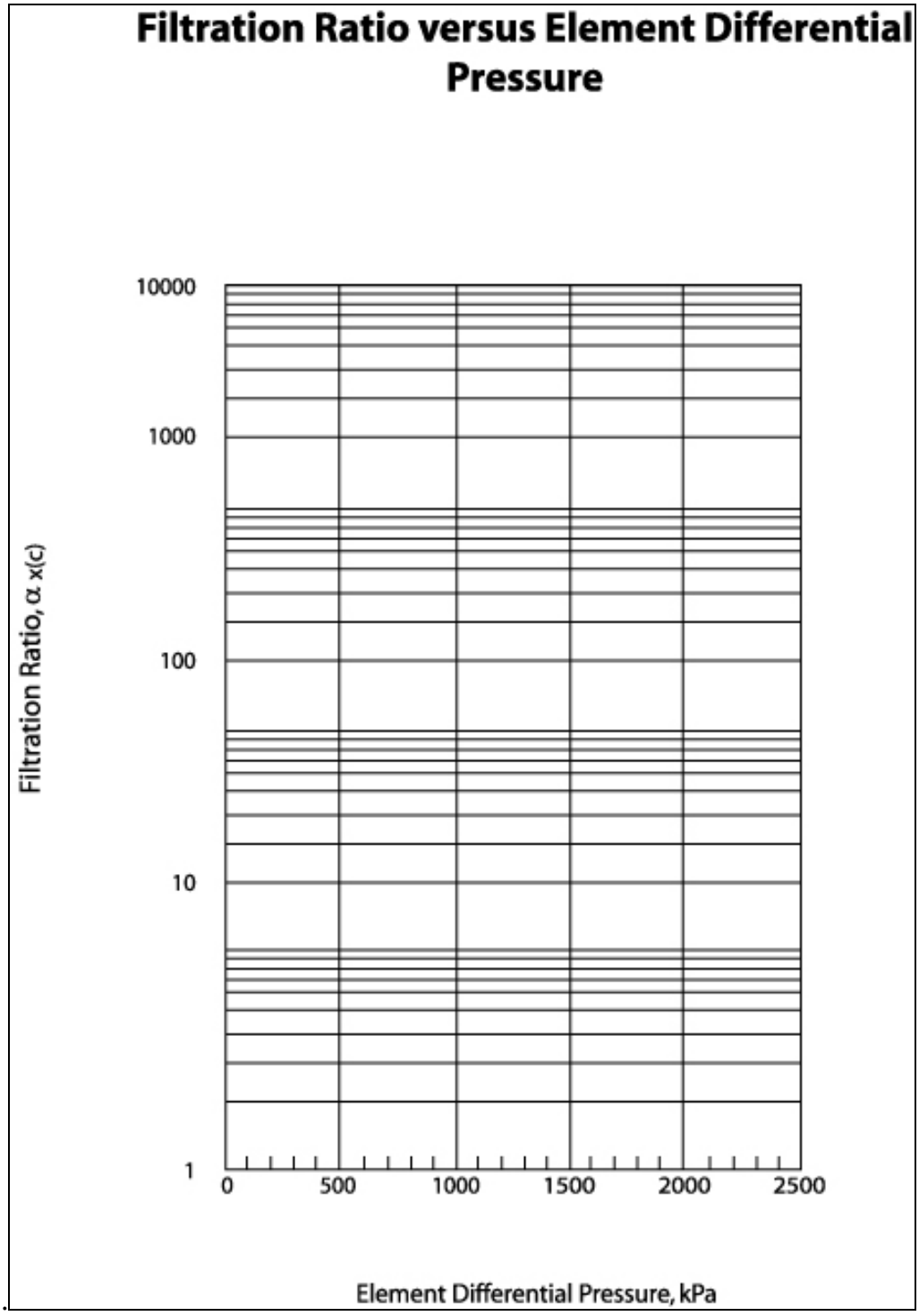


FIGURE B-13. Filtration Ratio Versus Element Differential Pressure

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APPENDIX B

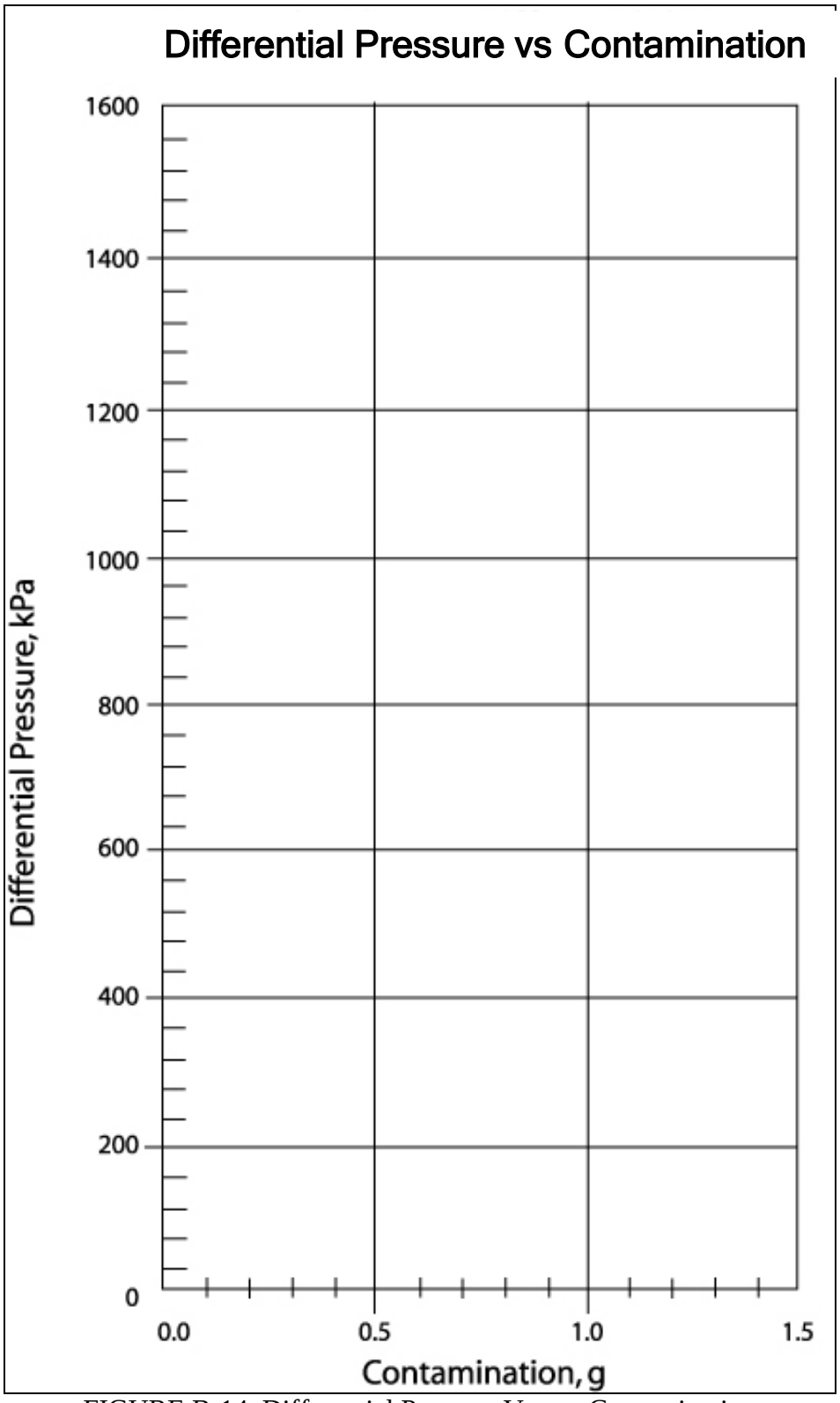


FIGURE B-14. Differential Pressure Versus Contamination

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APPENDIX B

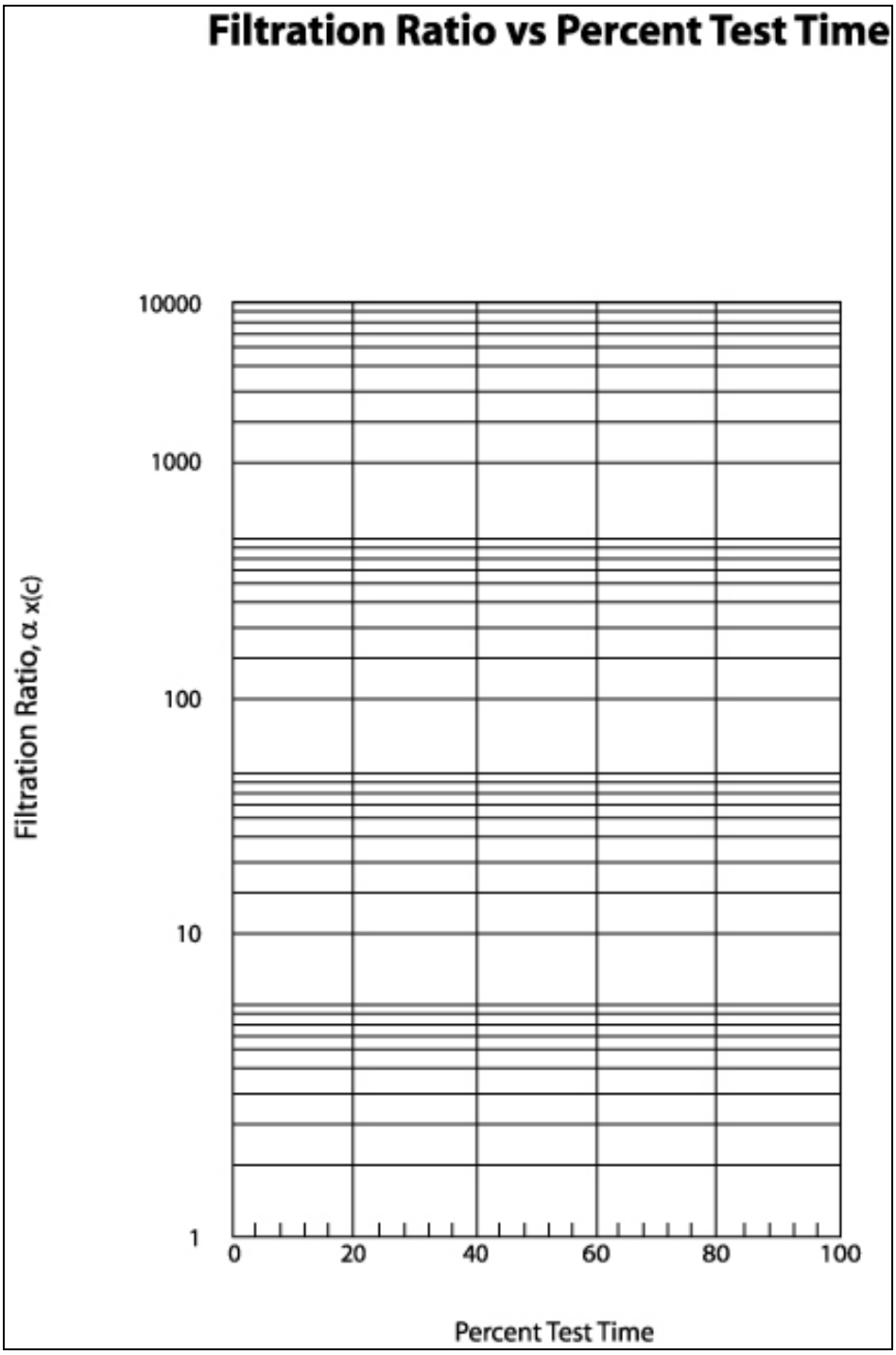
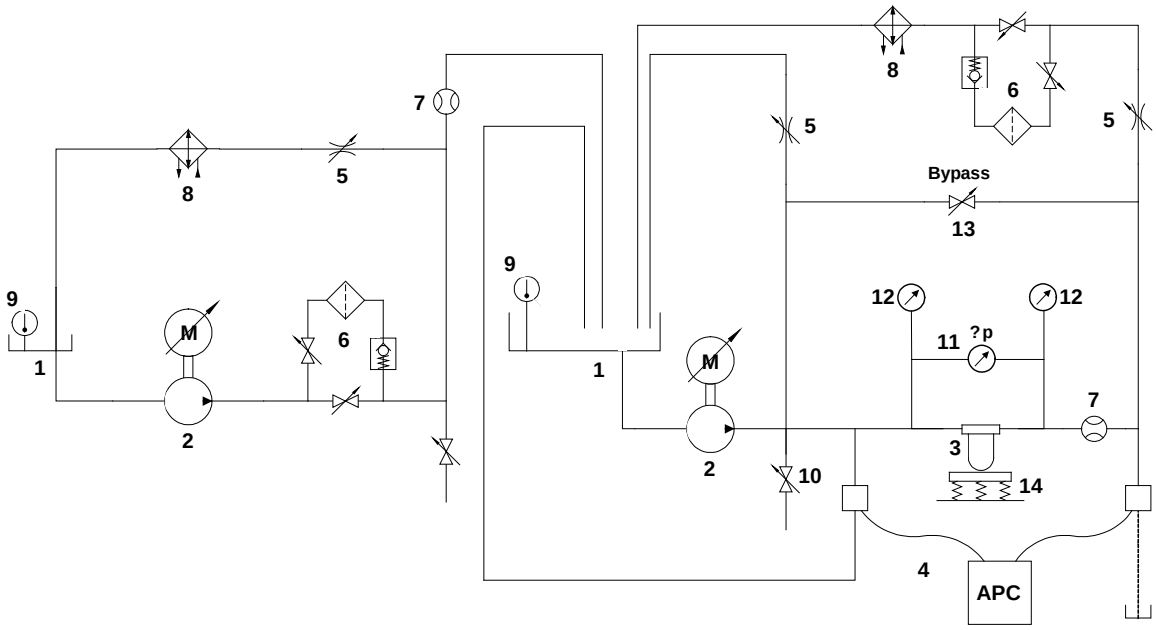


FIGURE B-15. Filtration Ratio Versus Percent Test Time

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APPENDIX C



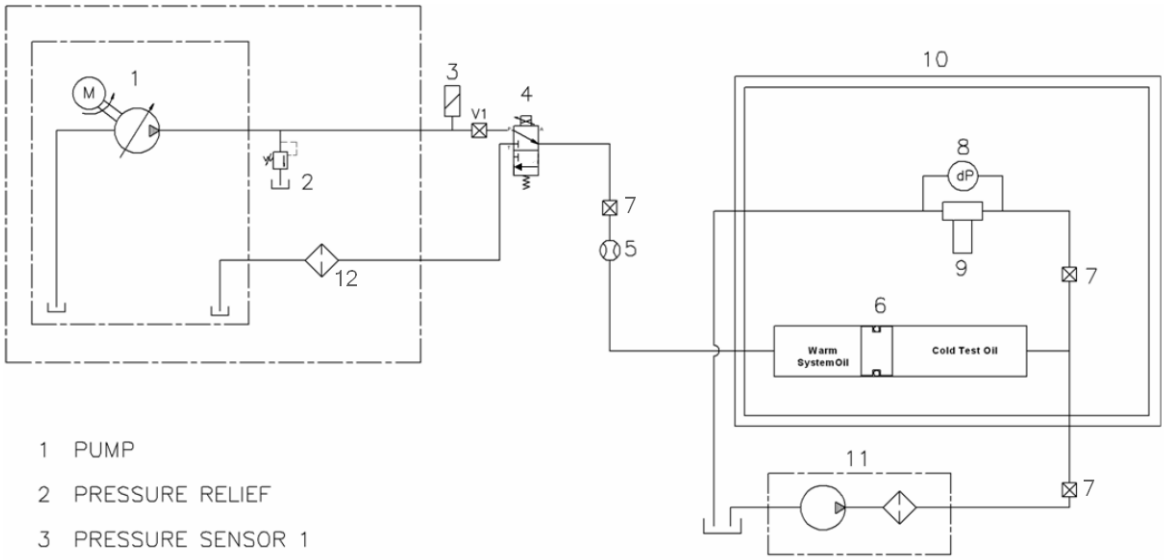
1	Reservoir	8	Heat Exchanger
2	Pump	9	Temperature Sensor
3	Unit Under Test	10	Sampling Valve
4	Particle Counting System	11	Differential Pressure Indicator
5	Regulating Valve	12	Pressure Gauge
6	Cleanup Filters	13	Cyclic Flow Bypass Valve
7	Flow Meter	14	Vibration Table

Note: The test set up shall use either a hydraulic pump with a variable speed motor (2) or a cyclic flow bypass valve (13). The option is at the discretion of the facility performing the test.

FIGURE C-1. Circuit Diagram of the Basic Test System

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APPENDIX C



1 PUMP
2 PRESSURE RELIEF
3 PRESSURE SENSOR 1
4 SWITCHING VALVE
5 FLOWMETER
6 PISTON ACCUMULATOR
7 VALVE
8 DELTA–P SENSOR
9 TEST ARTICLE
10 ENVIRONMENTAL CHAMBER
11 REFILL PUMP UNIT
12 FILTER

FIGURE C-2. Circuit Diagram for Cold Start Test

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APPENDIX D

1 SCOPE.

1.1 Scope. This appendix contains modifications to ISO 16889, Hydraulic Fluid Power – Filters – Multi-Pass Method for Evaluating Filtration Performance of a Filter Element (Second Edition 2008-06-15) that are applicable to filter elements provided in accordance with MIL-DTL-8815/31(AV).

2. APPLICABLE DOCUMENTS

INTERNATIONAL ORGANIZATION FOR STANDARDS (ISO)

ISO 16889	Hydraulic fluid power – Filters – Multi-pass method for evaluating filtration performance of a filter element

(Copies of these documents are available from the American National Standards Institute, 1819 L Street, NW, Washington, DC 20036, or online at http://www.ansi.org).

3. REQUIREMENTS

3.1 Exceptions to ISO 16889**.** ISO 16889 (Second Edition 2008-06-15) shall be used in its entirety except as indicated below:

Introduction. Replace the last paragraph on page 'v' with the following paragraph:

This test is intended to differentiate filter elements according to their functional performance, and it is also intended to represent performance in actual operation where the filter element is subjected to varying degrees of cyclic flow. The recommended flow cycle rate is identified Appendix A, FIGURE A-3.

1 Scope. Replace the first subparagraph with the following subparagraph:

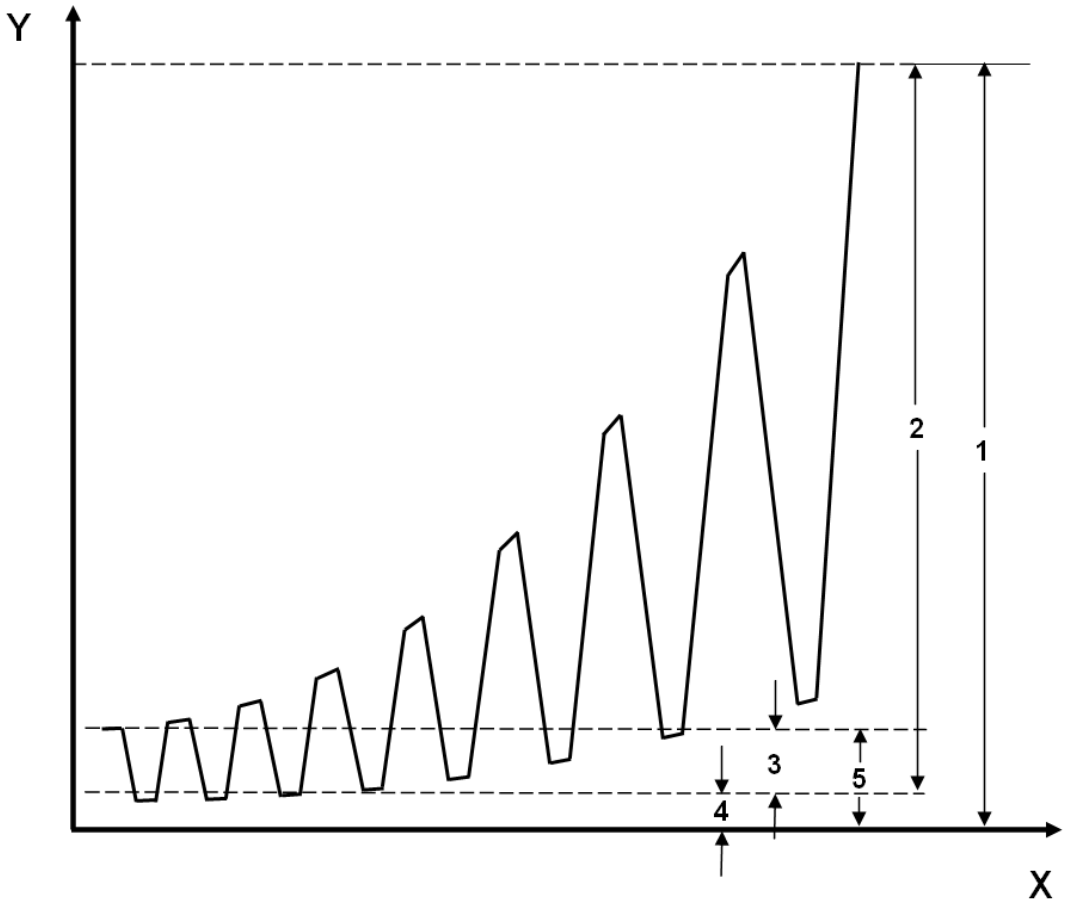
a) a multi-pass filtration performance test under cyclic flow conditions with continuous contaminant injection for hydraulic fluid power filter elements;

3 Terms and Definitions. Add the following paragraph and figure:

3.5
Cyclic Flow - change of flow from a specified flow rate to 25% of the rated flow rate at a specified waveform and a specified frequency.

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APPENDIX D



Key

X test time or mass injected
Y differential pressure
1 final assembly (end of test) differential pressure
2 terminal element differential pressure
3 clean element differential pressure
4 housing differential pressure
5 clean assembly differential pressure

Figure 1. Differential Pressure Conventions for Multi-Pass Test Under Cyclic Flow Conditions

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APPENDIX D

4 Symbols

Table 1 Letter Symbols. Add the following information to the table in the ISO 16889:2008(E):

Table I. Letter Symbols.

Symbol	Unit	Description or explanation
q_{min}	liters per minute	Minimum test flow rate (25% of q_{max})
q_{max}	liters per minute	Maximum test flow rate
t_R	second	Rise time in test flow rate cycle
t_F	second	Fall time in test flow rate cycle

Note. Replace all "*β*" with "α" in Table 1 as a reference for cyclic flow conditions**.**

6 Test Equipment. Add the following to paragraphs 6.4 and 6.7.1:

6.4 On-line counting system, and dilution system if necessary, validated IAW ISO 11943, and that synchronizes the initiation of a particle count with the "increase and decrease" of a test flow rate cycle as specified in Appendix A. No counting will be made during the transition area between high to low and low to high flow transition period.

6.7.1 Filter test system, consisting of the following:

g) a line that is equipped with an automatic control shut-off valve (for example: electrically-actuated ball valves or poppet type valves) capable of producing the required flow rate cycle at the designated frequency for the cyclic flow bypass.

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APPENDIX D

7 Accuracy of measurements and test conditions Replace Table 3 with the following:

Table III. Test Condition Value.

Parameter	Condition
Initial contamination level for filter test system	Less than 1% of the minimum level specified in Table 4, measured at the smallest particle size being counted.
Initial contamination level for injection system	Less than 1% of the injection gravimetric level.
Base upstream gravimetric level, mg/L [a]	3 ± 0.3
Recommended particle sizes for counting [b]	Minimum of five sized, including 30 µm(c), selected to cover the presumed filter performance range from $\alpha = 2$ to $\alpha = 1000$. Typical sizes are 4 µm(c), 5 µm(c), 6 µm(c), 7 µm(c), 8 µm(c), 10 µm(c), 12 µm(c), 14 µm(c), 20 µm(c), 25 µm(c), and 30 µm(c).
Sampling and counting method	On-line automatic particle counting.
Cyclic flow rate conditions	From q_{max} to q_{min} at a frequency in accordance with the waveform specified in Appendix A, Figure A-3.

[a] When comparing test results between two filters, the base upstream gravimetric levels should be the same.

[b] When a fine filter element is being tested, it might not be possible to count those particle sizes for which filtration ratios are low (for example, $\alpha = 2$ or $\alpha = 10$), and when a coarser filter element is being tested, it might not be possible to count or determine those particle sizes for which filtration ratios are high (e.g. $\alpha = 200$ or $\alpha = 1000$), because this can require measurements that are beyond the limits of the APC or the test conditions specified in this International Standard.

8 Filter performance test circuit validation procedures (replace the paragraphs 8.1.1 through 8.1.8, including Table 4 with the following)

8.1 Filter test system validation

8.1.1 Validate the filter test system at the minimum flow rate at which it is operated. Install a conduit in place of filter housing during validation. The conduit shall be selected so that it produces the maximum differential pressure expected during testing.

8.1.2 Validate the cyclic flow IAW the cyclic flow profile defined in FIGURE A-3 of MIL-DTL-8815/31(AV).

8.1.3 Validation shall be performed at the cyclic flow rate and test conditions specified in the paragraph 4.3 of MIL-DTL-8815/31(AV) specification document that include the minimum test flow rate (q_{min}) at which the filter test system will be operated. The minimum test flow rate (q_{min}) is 25% of the maximum test flow rate (q_{max}).

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8.1.4 Adjust the total fluid volume of the filter test system (exclusive of the clean-up filter circuit), such that it is numerically within the range of 25% to 50% of the maximum volume flow rate, expressed in liters per minute, with a minimum of 5 L.

It is recommended that the system be validated with a fluid volume numerically equal to 50% of the maximum test volume flow rate for flow rates less than or equal to 60 L/min, or 25% of the maximum test volume flow rates greater than 60 L/min.

NOTE This is the ratio of volume to flow rate required by the filter test procedure (see 10.3.4).

8.1.5 Contaminate the system fluid to the base upstream gravimetric level for the test condition as shown in Table 3 using the ISO 12103-A3 test dust.

8.1.6 Verify that the flow rate through each particle-counting sensor is equal to the value used for the particle-counter calibration within the limits of Table 2.

8.1.7 Circulate the fluid in the test system for 60 min, conducting continuous on-line automatic particle counts from the upstream sampling section for a period of 60 min. The sample flow from this section shall not be interrupted for the duration of the validation.

8.1.8 Record the cumulative on-line particle counts at equal time intervals not exceeding 1 min for the duration of the 60 min test at the particle sizes selected from those given in Table 3, including the 30 μm(c) particle size.

8.1.9 Accept the validation test only if:

a) the particle count obtained for a given size at each sample interval does not deviate more than 15% from the average particle count from all sample intervals for that size, and

b) the average of all cumulative particle counts per milliliter is within the range of acceptable counts shown in Table IV.

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Table IV. Particle size versus acceptable cumulative particle counts per milliliter.

Particle size	**Acceptable cumulative particle counts per milliliter** [a]	
	Test condition (3 mg/L)	
µm(c)	min.	max.
1	104 000	128 000
2	26 100	31 900
3	10 800	13 200
4	5 870	7 190
5	3 590	4 390
6	2 300	2 830
7	1 510	1 860
8	1 010	1 250
10	489	609
12	265	335
14	160	205
20	46	64
25	16	27
30	6	12
40	1.1	4.5
50	0.15	2.4

[a] The minimum and maximum values are based on particle counts determined by the National Institute for Standards and Technology (United States) for SRM 2806 (see ISO/TR 16144) with a calculated variation based on the Poisson distribution.

8.1.10 Validate the on-line particle counting system, , in accordance with ISO 11943 while the filter test system is under cyclic flow conditions, and report validation test results in the format shown in FIGURES B-1 through B-4 of MIL-DTL-8815/31(AV).

8.2 Validation of contaminant injection system. Replace paragraphs 8.2.1 and 8.2.4 with the following:

8.2.1 Validate the contaminant injection system at the maximum gravimetric level, maximum injection system volume, average injection flow rate, and for the length of time required to deplete the complete usable volume.

8.2.4 Initiate injection flow from the contaminant injection system, collecting this flow external from the system. Obtain initial sample at this point and measure the injection flow rate.

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10 Preliminary preparation Replace paragraphs 10.2 and 10.3 with the following:

10.2 Contaminant Injection System. A turbulent means should be provided for transferring fluid from the contaminant injection system to the filter test system. The flow rate, q (i), shall be maintained constant at one flow rate throughout the test duration and it is suggested to be maintained in the range 100 and 250 ml/min depending on the test system requirements. Refer to the ISO 16889 for validation requirements of the gravimetric system.

10.3 Filter test system. Add the following to the WARNING sentence, and replace paragraphs 10.3.3 and 10.3.4 with the following:

WARNING Use of an anti-static agent that has a date code older than 18 months is not recommended.

10.3.3 Circulate the fluid in the filter test system at maximum rated flow rate and at a test temperature at 160° F +/- 5° F and record the temperature and differential pressure of the empty filter housing in accordance with ISO 3968.

10.3.4 Adjust the total fluid volume of the filter test system (exclusive of the clean-up filter circuit) such that its value in liters is numerically between 25% and 50% of the designated test volume flow rate through the filter, expressed in liters per minute, with a minimum value of 5 L.

If the designated test volume flow rate is less than or equal to 60 L/min, it is recommended that the filter test system fluid volume be numerically equal to 50% of the maximum test volume flow rate. If the designated test volume flow rate is greater than 60 L/min, it is recommended that the filter test system fluid volume be numerically equal to 25% of the maximum test volume flow rate.
NOTE Repeatable results require that the system volume be maintained constant. The specified range of ratios between the test system fluid volume and the test volume flow rate from 1:4 to 1:2 minimizes the physical size of the system reservoir, as well as the quantity of test fluid required, while maximizing the mixing conditions in the reservoir.

11 Filter performance test. Replace paragraphs 11.8 through 11.12 with the following and add paragraph 11.13:

11.8 Operate the cycling valve in the test system to generate and repeat the flow rate cycle shown in FIGURE A-3.

a) t_R (rise time) = 150 ms – 250 ms

b) t_F (fall time) = 150 ms – 250 ms

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11.9 Initiate the filter test by

a) allowing the injection flow synchronized with the cyclic test flow rate to enter the filter test system reservoir. The average injection flow rate is the average of maximum and minimum injection flow rate.

b) starting the timer, and

c) diverting the downstream sample flow from the test system to maintain a constant system volume within a tolerance of ± 5% [see 10.3.6 a)].

11.10 Conduct and record the on-line particle counts on the upstream and downstream fluid at equal time intervals not exceeding 1 min until the differential pressure across the filter assembly at maximum test flow rate has increased to the terminal value calculated in 11.3. Synchronize the initiation of the particle count as shown in FIGURE A-3 of MIL-DTL-8815/31(AV).

The upstream and downstream sensor flow rates should be equal to the values chosen in 10.3.6 b), within the limits specified in Table 2.

Flow rates through sensors should be monitored and recorded throughout the test and maintained within the limits specified in Table 2.

Care should be taken to use on-line dilution as required to avoid exceeding the coincidence limit of the automatic particle counter, as determined in accordance with ISO 11171.

It is recommended that the flow rate and dilution ratio be controlled and recorded to calculate the exact amount of test fluid that is passed through the sensor for each count.

It is recommended that a minimum counting volume of 10 mL be used to obtain statistically significant particle counts.

11.11 Record the assembly differential pressure at the beginning of each particle count throughout the test.

Continuous differential pressure measurements using a differential pressure transducer are recommended for this purpose.

11.12 Extract a bottle sample for gravimetric analysis from upstream of the test filter when the assembly differential pressure has reached 80% of the terminal assembly differential pressure.

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11.13 Conclude the test at the final assembly differential pressure at the maximum test flow rate by:

a) recording the final test time,

b) diverting the injection flow from the filter test system,

c) stopping the flow rate cycle, and

d) stopping the flow to the test filter.

12 Calculations. Add the following subparagraphs to the paragraph 12.8.

12.8.1 Compute the 'High to Low Average' filtration ratio by using the equation in FIGURE A-3 of the MIL-DTL-8815/31(AV).

12.8.2 Compute the 'Low to High Average' filtration ratio by using the equation in FIGURE A-3 of the MIL-DTL-8815/31(AV).

12.8.3 Compute the average performance of three filter elements. In order to compute the average of three different test results perform the following: convert the Alpha Grand Average Ratio and the AFR ratio to filter efficiency. Average efficiency averages, and convert back to the overall average of all three filter elements. Apparent and retain capacities can be computed by using the straight averages.

NOTE The (c) is the subscript that signifies the particle counters are calibrated to ISO 11171.

13 Data presentation. Replace paragraph 13.5 and NOTE 2 with the following:

13.5 Plot on semi-log (log linear) coordinates average α versus particle size, *x*, with the α values on the log scale and α = 10 000 as the highest value plotted (see the example in C.3).

NOTE 2 For α values greater than 10 000, use the value of 10 000 in Equation (16).

14 Identification statement. Replace the paragraph with the following:

It is strongly recommended to manufacturers who have chosen to conform to this International Standard that the following statement be used in test reports, catalogues and sales literature:

"Method for determining filtration performance of a hydraulic filter element under cyclic flow conditions conforms to Military Standard MIL-DTL-8815/31(AV), *FILTER ELEMENTS, NON-CLEANABLE, FLUID PRESSURE, HYDRAULIC LINE, 3,000 PSI, 5 MICRON ABSOLUTE, 65°F TO 275°F*."

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APPENDIX D

Figure 2 Replace Figure 2 with FIGURES B-6 and B-7 of MIL-DTL-8815/31(AV).

Figure B.1 Replace Figure B.1 with FIGURE C-1 of MIL-DTL-8815/31(AV).

B.2.2 Filter test system. Add the following paragraph as paragraph B.2.2.7 to this section:

B.2.2.7 If a variable speed pump is not used, the cyclic flow bypass valve section is the key to performing the cyclic flow multi-pass test. The valve will be required to regulate the flow bypassing the test filter. This valve controls the bypass of fluid around the test filter at the specified cycle rate. Upstream of the test filter it should be connected between the pump and the APC sample port. Downstream of the test filter it should be connected between the APC sample port and the counter pressure regulation valve. Ball, diaphragm, weir or pinch valves with automatic actuation are suitable for this purpose.

B.2.3.3 Pump. Add the following at the end of paragraph B.2.3.3:

When using centrifugal pumps, vertical mounting with the inlet down or horizontal mounting with the discharge at the bottom have proven successful.

Preparing Activity:
Army- AV

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